

1. Administrative & Study Identification

Full Study Title: A Single-blind, Randomised, Placebo-controlled Study to Investigate the Safety, Tolerability, and Pharmacokinetics of Inhaled AZD4604 Following Single Ascending and Multiple Doses in Healthy Japanese and Chinese Participants
Short Title/Acronym: AZD4604 Phase 1 in Healthy Japanese and Chinese Participants **Protocol Number:** D8210C00002 **NCT Number:** NCT06519968 **Sponsor Name:** AstraZeneca **Investigational Product:** AZD4604 (londamocitinib, an inhaled Janus kinase 1 [JAK1] inhibitor) **Phase of Development:** Phase 1 **Medical Monitor:** AstraZeneca Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To investigate the safety and tolerability of inhaled AZD4604 when administered as single or multiple inhaled doses to healthy Japanese and Chinese participants. **Secondary Objectives:** To investigate the pharmacokinetic (PK) parameters of AZD4604 (e.g., Cmax, AUC) across single ascending dose (SAD) and multiple dose (MAD) cohorts.

2.2 Background & Rationale Janus kinase 1 (JAK1) is a promising target for asthma treatment, as it may address inflammation uncontrolled by inhaled corticosteroids. Londamocitinib (AZD4604) is a selective JAK1 inhibitor designed for inhaled delivery. This Phase 1 study was conducted to establish the safety, tolerability, and pharmacokinetic characteristics of the drug in healthy Japanese and Chinese participants to support their inclusion in further global clinical development programs for asthma.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Placebo-controlled **Blinding:** Single-blind **Randomization:** Randomised

3.2 Planned Enrollment & Duration **Target N\$:** 56 (Actual Enrollment: 56 healthy participants) **Number of Sites:** Phase 1 Clinical Research Unit(s) **Estimated Study Start:** 15-Jul-2024 (Actual) **Estimated Primary Completion:** 27-Dec-2024 (Actual)

4. Subject Selection (Eligibility Criteria)

- 4.1 Inclusion Criteria**
1. Healthy Japanese and Chinese male or female participants, aged 18 to 55 years.
 2. Body mass index (BMI) between 18 and 30 kg/m² and weight of at least 45 kg.
 3. Forced Expiratory Volume at first breath (FEV₁) $\geq 80\%$ of the predicted value in accordance with ATS/ERS criteria at screening and admission.
- 4.2 Exclusion Criteria**
1. History or presence of clinically important disease (e.g., gastrointestinal, hepatic, renal, or abnormal immune function) which may put the participant at risk.
 2. Current smokers, use of nicotine products within the previous 6 months, or a smoking history of > 5 pack-years.
 3. Increased risk of infection, or receipt of any vaccine 30 days prior to the first dose.
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5. Intervention & Treatment Plan

- 5.1 Investigational Regimen Dosage/Frequency:**
- * **SAD Cohorts (Part 1a & 2a):** A single ascending dose on Day 1.
 - * **MAD Cohorts (Part 1b & 2b):** Twice daily (BID) dosing from Day 1 to Day 6, followed by a single final dose on Day 7.
- Route of Administration:** Inhalation via Dry Powder Inhaler (DPI). **Duration of Treatment:** 1 day of exposure for SAD cohorts; 7 days of exposure for MAD cohorts.
- 5.2 Concomitant & Prohibited Therapy Permitted:** Standard allowed medications for Phase 1 healthy volunteer studies, evaluated case-by-case by the investigator. **Prohibited:** Any other investigational drugs, vaccines within 30 days of the first dose, and nicotine or alcohol consumption.
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6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to -1	Informed Consent, Medical History, BMI, FEV ₁ Labs
Baseline	Day 1	Randomization, First Dose, PK Sampling Initiation
Interim	Day 2 to 6	Safety Labs, Vital Signs, Continuous PK Monitoring, BID Dosing (MAD Cohorts)
End of Study	Day 7 (SAD) or Day 13 (MAD)	Final Safety Assessments, Exit PK Sampling, Discharge from Clinical Unit

7. Outcome Measures (Endpoints)

- 7.1 Primary Endpoint Definition:** Safety and tolerability endpoints, evaluated via the incidence of adverse events (AEs), serious adverse events (SAEs), and vital sign/laboratory abnormalities over the study period. **Measurement Method:**

Continuous clinical observation and standard laboratory analysis during confinement.

7.2 Secondary & Exploratory Endpoints Pharmacokinetic (PK) parameters, including maximum observed plasma concentration (\$C_{max}\$) and area under the plasma concentration-time curve (\$AUC\$). Evaluation of drug accumulation (steady-state accumulation ratio) with BID dosing.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Standard collection via daily clinical observation and participant reporting during confinement. **Serious Adverse Events (SAEs):** Required to be reported within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** Internal safety review process utilized to review data before dose escalation from SAD to MAD cohorts.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Analysis Set (all participants receiving \$\ge 1\$ dose) and PK Analysis Set. **Sample Size Justification:** \$N=56\$ was chosen based on typical Phase 1 study standards to adequately characterize the PK and initial safety profile in specific ethnic populations, rather than formal statistical power calculations. **Handling of Missing Data:** No imputation for missing safety data; standard non-compartmental analysis rules applied for missing PK time points.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** High-frequency onsite monitoring and 100% Source Document Verification (SDV) typical for Phase 1 clinical confinement units. **Audit Trail:** Strict eCRF locking, tracking of protocol deviations, and PK sample chain-of-custody tracking.

11. Study Identifiers & Governance

Primary ID: D8210C00002 **Registry Number:** NCT06519968 **Sponsor/Lead Organization:** AstraZeneca **Study Title:** AZD4604 Phase 1 in Healthy Japanese and Chinese Participants **Official Regulatory Title:** A Single-blind, Randomised, Placebo-controlled Study to Investigate the Safety, Tolerability, and Pharmacokinetics of Inhaled AZD4604 Following Single Ascending and Multiple Doses in Healthy Japanese and Chinese Participants

12. Clinical Framework (Asthma & Healthy Volunteer Specific)

Primary Condition: Healthy Participants (Target indication: Asthma) **Diagnosis Threshold:** Normal lung function ($FEV_1 \geq 80\%$ predicted) **Medical Methodology:** Phase 1 Pharmacokinetics and Safety Assessment **Optimization Target:** Target steady-state PK profiling and tolerability assessment

13. Study Procedures & Methodology

Management Pathway: Phase 1 Clinical Research Unit Confinement **Education Protocol (HCP):** Protocol specific training & PK sampling guidelines **Education Protocol (Patient):** Dry Powder Inhaler (DPI) device use instruction and compliance monitoring **Quality Audit Mechanism:** Clinical site monitoring and PK sample integrity tracking

14. Observation & Follow-Up Schedule

Total Duration: Up to 41 days (including up to 28 days screening and 13 days evaluation) **Onsite Visit Cadence:** Confinement from Day -1 through Day 7 (SAD) or Day 13 (MAD) **Remote Monitoring Frequency:** N/A (In-patient study methodology) **Checklist Review Interval:** Daily AE and vital signs review during the active dosing period

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]			Lead	
Statistician	[Name]	_____	[DD-MMM-YYYY]				